nonchalantly, a few months later, “but I think we’re getting better at realizing one of the things we’ve been saying in string theory for a long time, which is that spacetime isn’t fundamental.”

Oh, yeah. That small detail. Sure.

Johnson’s approach to the question is a bit different. There are some intriguing hints in quantum gravity theories of unexpected connections between physics on small and large scales, in ways that don’t make sense in our usual thinking about how spacetime works. A simplified explanation might be that if you imagine doing experiments in a hypothetical kind of space that has a certain radius, let’s call it R, the results of that experiment would look exactly like the results of the same experiments in a much smaller space, with a radius equal to 1 divided by R. In string theory, this is called T-duality, and it’s a weird enough coincidence that it seems like it *has* to be telling us something deep. “If you ask people about this question,” Johnson says, “the answer that people would give is that in some sense, none of it is real. In the sense that by undermining large and small, what you’re really doing is undermining the whole business of spacetime in the first place.”

Some theorists have tried to reassure me. Sean Carroll, a cosmologist at Caltech who’s interested these days in the underpinnings of quantum mechanics, thinks we are all being a little rash to dismiss spacetime as not strictly real. “It’s *real* but not *fundamental*,” he tells me. “Just like this table is real but not fundamental. It’s a higher level of emergent description. That doesn’t mean it’s not real.” Basically we shouldn’t get too hung up on this because it’s not like spacetime isn’t there, it’s just that if we really understood what it was made of, it would look, at a deeper level, like something else entirely.

This does not, in fact, reassure me.[fn7] As a physicist I always try to maintain some level of dispassionate reserve when it comes to my subject, but the notion that spacetime is only real in the sense that it is something we can talk about and sit on but not in the sense of being what the universe is *actually made of* still makes me feel like it might collapse underneath me at any moment.

Whether or not this has relevance to how or when the universe will end is still an open question. However real spacetime is or isn’t, we all live there, and what happens to spacetime is bound to affect us. But if thinking about emergent spacetime or new formulations of quantum mechanics leads us to